

Foundations: What is Data Science & Data Analytics

Read this first. It explains the ideas behind Week 1 — no code yet. Basahin muna ito bago ang mga activity.

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1. Data vs. Information

Data are raw facts — numbers, text, dates — with no meaning yet on their own (e.g. 23, 61, "Sunny"). **Information** is data that has been organized and interpreted so it answers a question (e.g. "The average temperature across our stations is 18°C").

*Tagalog: Ang **data** ay hilaw na impormasyon — mga numero o salita na wala pang kahulugan. Nagiging **information** ito kapag inayos at binigyang-kahulugan para sagutin ang isang tanong.*

Key idea: a data analyst's job is to turn data into information someone can act on.

2. The data workflow

Almost every data project follows the same loop. You will do each step in Week 1:

1. **Collect** — gather the data (a survey/form, a web page, a sensor, a file).
2. **Clean** — fix missing values, remove duplicates, correct types.
3. **Explore & analyze** — summarize, group, and compare.
4. **Visualize** — turn numbers into charts people can read.
5. **Interpret** — say what it means and what to do next.

Tagalog: halos lahat ng proyekto ay sumusunod sa parehong hakbang — kolekta → linis → suri → i-chart → bigyang-kahulugan.

3. What is EDA (Exploratory Data Analysis)?

EDA is the "get to know your data" step — before any conclusions. You look at how many rows and columns you have, what types they are, the min/max/average of the numbers, how categories split, and whether two things move together (**correlation**). EDA is about *asking questions*, not proving a point.

Remember: a correlation near **0** means "no clear relationship" — and reporting that honestly is just as valuable as finding a strong one.

4. Why cleaning matters

Real collected data is messy: blank cells, duplicate rows, wrong formats. If you analyze dirty data you get **wrong answers that look convincing**. A first-pass clean — drop duplicates, fill missing numbers with the average, fill missing text with the most common value — makes the analysis trustworthy.

Tagalog: kung marumi ang data, mali ang lalabas na sagot kahit maganda tingnan. Kaya naglilinis muna bago mag-analyze.

One more honest rule: always say whether your data is **real or simulated**. In this class the datasets are **simulated sample data** for training — the *skills* are real, the numbers are invented.

Proceed to the activities →